

Anchoring Dynamics in Mobile Augmented Reality: Balancing Accessibility and Visual Interference

Category: Research

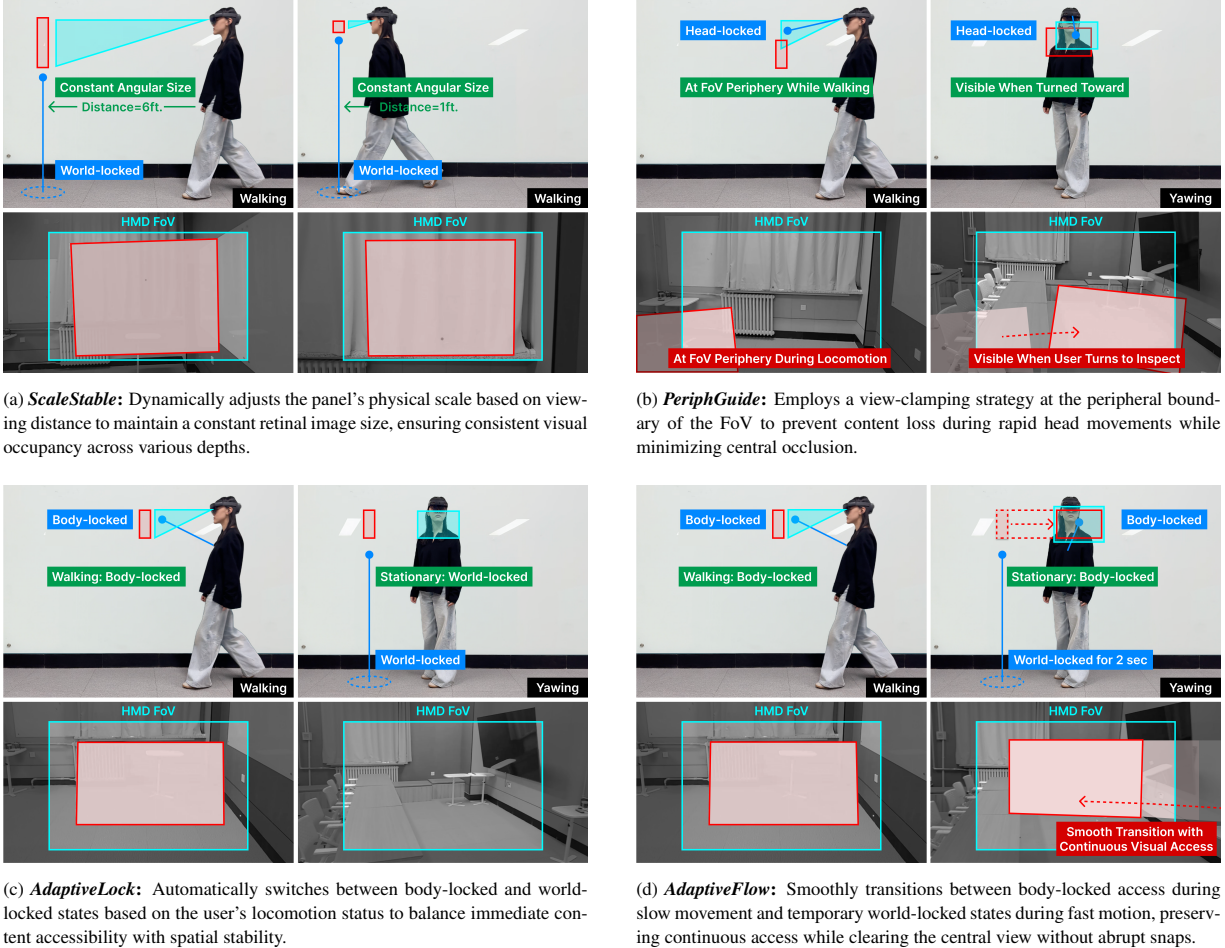


Figure 1: Our four anchoring techniques for mobile multitasking with AR HMDs.

ABSTRACT

Augmented Reality (AR) overlays digital information onto physical environments, with anchoring techniques managing virtual content placement relative to user reference frames. Common approaches—world-locked (stable to environment), head-locked (tied to gaze), and body-locked (torso-relative)—suffice for static scenarios but falter in mobile AR, where users juggle navigation, dynamic surroundings, and cognitive tasks amid competing visual demands. This paper explores trade-offs between interface accessibility and field-of-view interference across mobility and cognitive loads. We introduce four techniques extending traditional anchoring: **ScaleStable** (world-locked constant angular size), **PeriphGuide** (head-locked peripheral clamping during locomotion), **AdaptiveLock** (stationary world-locked, walking body-locked), and **AdaptiveFlow** (body-locked with smooth world-locked transitions in fast motion). Two studies simulated mobile AR tasks: *Study 1* (individual dual-tasking, high mobility) favored adaptive/body-locked methods for access and recall; *Study 2* (collaborative sense-making, high cognition) highlighted **ScaleStable**'s stability for collaboration. Findings underscore context-aware anchoring, assessing full visual-cognitive demands over single-factor optimization.

Index Terms: Augmented Reality, Anchoring Techniques, Collaboration, Dynamic Meeting, Attention, Multitasking, OST-HMD.

1 INTRODUCTION

Augmented Reality overlays digital information onto physical environments to create seamless hybrid experiences that enhance users' understanding and interaction with their surroundings [38, 2]. Central to this integration are anchoring techniques, which manage the spatial relationship between virtual content and the user's reference frames [4]. Three approaches are commonly used in practice: (1) **world-locked**, which fixes content to physical locations, providing stable environmental context but requiring user movement to maintain access [48, 33]; (2) **head-locked**, which attaches content to the user's view direction, ensuring continuous availability but occluding parts of the visual field [47, 25]; and (3) **body-locked** anchoring, which positions content relative to the user's torso orientation, balancing accessibility with reduced central occlusion [17, 33, 25].

While these techniques are simple to understand and work well in many scenarios, they prove less sufficient for the complex demands under mobile AR usage [21]. In mobile contexts, users often need to simultaneously process both physical and digital in-

formation while navigating dynamic environments, creating competing demands for visual attention and cognitive resources [35]. The challenge intensifies as task complexity and mobility demands increase: users must dynamically shift attention between different physical objects, navigate through entirely changing surroundings, and maintain access to digital information throughout continuous movement [4]. Traditional anchoring approaches—optimized for either stability (world-locking) or digital information accessibility (head/body-locking)—can become insufficient to adapt to these fluctuating priorities.

Rather than adhering to a single universal policy, optimal anchoring strategy design can depend on contextual factors including user mobility state, environmental complexity, task cognitive load, and attention switching frequency. This paper investigates the fundamental trade-offs between interface accessibility and field-of-view interference in mobile AR contexts, examining how four anchoring strategies affect user performance across varying cognitive and mobility demands. We developed four techniques that expand beyond basic world/head/body locks: *ScaleStable* (world-locked with constant angular size), *PeriphGuide* (head-locked peripheral clamping during locomotion), *AdaptiveLock* (toggles between world-locked when stationary and body-locked while walking), and *Adaptive-Flow* (body-locked with smooth brief world-locked transitions during fast motion).

Through two controlled studies simulating realistic mobile AR usage scenarios, we found important trade-offs in technique preference and effectiveness. *Study 1* (individual dual-tasking under basic mobility demands) showed adaptive and body-locked techniques *AdaptiveLock* outperformed *ScaleStable* on interface access and memory recall, suggesting accessibility matters most when cognitive demands are moderate but mobility requirements are high. However, *Study 2* (collaborative sensemaking under higher cognitive load but moderate mobility requirements) revealed *ScaleStable*’s stability advantages outweighed accessibility costs, yielding better perceived collaboration despite lower interface attention.

These findings suggest that future AR systems can consider employing dynamic anchoring strategies, optimizing the interference-accessibility-disruption balance through real-time assessments of task demands rather than relying solely on motion-based heuristics. Key enablers can include monitoring environmental complexity, detecting cognitive load, analyzing mobility patterns, recognizing task phases, and tracking user attention states. Ultimately, effective AR interfaces will emerge from intelligent, context-aware systems that holistically evaluate user needs, environmental factors, and cognitive states to dynamically select optimal strategies.

2 RELATED WORK

2.1 Remote Collaborative AR Systems

Effective remote Augmented Reality (AR) collaboration requires mechanisms to convey presence, coordinate attention, and establish shared spatial references [41, 12]. For example, remote experts frequently guide on-site workers via persistent spatial annotations [41]. Early systems achieved this primarily through avatars and gaze visualizations [12]. Later research demonstrated that world-locked spatial references outperform head-locked, especially during less-dynamic problem-solving tasks [8, 10].

However, foundational AR research predominantly assumes static users. Recent studies on situated collaboration reveal that walking and environmental transitions complicate coordination mechanisms designed for stationary contexts [46]. Locomotion introduces dual-task demands: users must simultaneously process remote partner cues and navigate physical surroundings [4, 20]. Consequently, managing virtual information panels while concurrently navigating physical environments introduces distributed attention

demands that remain underexplored in collaborative mobile contexts.

2.2 Multitasking and Attention Management in Augmented Reality

By overlaying virtual content onto the physical world, AR creates complex multitasking scenarios. Dual-task interference research [36, 24] confirms that combining physical and virtual tasks creates resource competition, degrading performance [6]. This interference intensifies during mobility, as walking consumes attentional resources that compete with AR content processing [27].

Unlike mobile phones, which allow discrete attention shifts, AR head-mounted displays (HMDs) induce sustained attentional fragmentation by inserting content into the continuous field of view [21, 39]. Furthermore, task-switching between environmental navigation and AR interaction incurs substantial cognitive overhead [21]. While persistently visible AR content may reduce explicit reorientation costs, it generates continuous low-level visual interference [1]. This underscores the critical need for optimal AR content placement to facilitate efficient attention recovery and resumption.

2.3 Anchoring Techniques for AR Content

AR content is typically anchored via three reference frames [36]: *world-locked* (fixed to the environment), *body-locked* (fixed to the user’s torso), and *head-locked* (fixed to the head orientation). World-locked content preserves spatial stability, making it ideal for navigation [48], yet it introduces viewpoint dependencies that require physical movement to access out-of-view information [33]. Conversely, head-locked content ensures constant visibility for quick-access tasks [25] but risks visual clutter and lacks spatial grounding [47]. Body-locked content offers a compromise for sustained interaction [17] but introduces tracking complexities [33].

While prior studies identify task-specific preferences, most evaluate static, single-user scenarios. Although recent adaptive systems [26, 36] explore mobile AR, they focus on individual workload rather than collaborative coordination. In remote AR collaboration, locomotion and shared coordination demands jointly can dictate optimal anchoring. Our study addresses this by investigating how these techniques should adapt during locomotion to optimize attention in dynamic, collaborative multitasking scenarios.

3 STUDY OVERVIEW

To identify the limitations of current anchoring techniques in mobile AR multitasking, we began with a formative pilot study ($N=12$). The results highlighted inherent trade-offs in static anchoring, particularly between field-of-view interference and information accessibility, that intensify under mobility and varying cognitive demands. These insights motivated the design of four novel adaptive techniques that address the interference-accessibility balance through different spatial strategies. We evaluated these techniques by decoupling individual and collaborative challenges across two sequential studies: *Study 1* ($N = 12$) measured single-user attention allocation and task performance during mobile information encoding, while *Study 2* ($N = 24$, 12 dyads) assessed dyadic attention coordination and collaboration quality in a remote investigative scenario.

Participants. All studies complied with our university’s human research ethics policy; informed consent was obtained from all participants, and no participant took part in more than one study. Participants were volunteers from the local university community and were compensated at $\approx$ USD 6 per hour, consistent with local standards. All had normal or corrected-to-normal vision, and were AR HMD novices, to ensure comparable perception of AR content.

4 PILOT STUDY

To understand user challenges with existing anchoring techniques (world-, body-, and head-locked) during mobile multitasking with AR HMDs, we conducted a pilot study (N=12) in which participants (Sec. 3) performed a walking—reading—communicating task (see Appendix Sec. A for details).

4.1 The Anchoring Trade-off

Consistent with prior findings [34, 47, 26], our pilot study revealed mobile AR anchoring involves choosing between two competing strategies: **world-based anchoring**, which has content fixed in physical space, and **user-based anchoring**, which has content attached to the user’s head (head-lock) or torso (body-lock). The effectiveness of these strategies seem to relate to a number of factors below.

Field-of-view interference. This refers to virtual elements obstructing or disrupting the user’s natural perception of the real-world environment within their visual field. Body-based anchors keep content in the central visual field, providing immediate access but persistently occluding the physical environment[24]. World-based anchors reduce this occlusion by placing content at dedicated locations, but require deliberate gaze redirection to access.

Information accessibility. This refers to whether the information presented by the technique can be accessed by users. Body-based anchors remain available regardless of movement. World-based anchors become progressively harder to locate, may leave the field of view, and can grow illegible at distance.

Mobility shifts the balance between these factors. When stationary, users can afford to seek out world-anchored content, benefiting from reduced occlusion. During movement, environmental processing demands increase while cognitive bandwidth for locating distant content decreases—world-anchored content becomes not only less accessible but actively disruptive. Body-anchored content stays accessible but occludes the environment precisely when awareness is most critical. Traditional approaches must sacrifice one regime for the other, motivating us to explore adaptive techniques that adjust spatial behavior based on user state.

5 ADAPTIVE ANCHORING DESIGN

Based on the trade-offs identified above, we developed four adaptive anchoring techniques that balance field-of-view interference and information accessibility in different ways:

5.1 Four Anchoring Techniques

5.1.1 Static Anchoring

The following two techniques utilize fixed geometric rules to enhance either legibility or access:

ScaleStable (Fig. 1a): *ScaleStable* is a world-locked technique preserving constant apparent panel size via constant visual angle as user–panel distance changes. Fixed to an environmental anchor (0.5–3.5 m range); scaling up when distant for legibility, down when close to avoid occlusion. It continuously faces the user for readability [10] and applies temporal smoothing (Move/Rotate Lerp = 0.1) to reduce jitter [45].

PeriphGuide (Fig. 1b): *PeriphGuide* is a head-locked, availability-first technique keeping the panel near the line of sight but off foveal center. Rendered at 0.8 m (clamped 0.8–1.0 m) with per-frame smoothing (Lerp = 0.1), its pose uses fixed angular offset clamped $\pm 40^\circ$ H / $\pm 30^\circ$ V inside HMD’s FOV ($\approx 52^\circ$ diag., 43° H, 29° V) [29]; reorients if outside bounds. Like Tag-along but peripheral—recoverable by small head turn—limiting central motion for reading stability [43].

5.1.2 Adaptive Anchoring

Our adaptive techniques (*AdaptiveLock* and *AdaptiveFlow*) classify motion via thresholds on walking speed—**stationary** (< 0.3 m/s), **slow-moderate** (0.3–1.5 m/s), **fast** (> 1.5 m/s)—and head yaw ($> 150^\circ$ /s), per gait/vestibular standards [30]. Exceeding thresholds triggers transitions; lower speeds preserve stability.

AdaptiveLock (Fig. 1c): *AdaptiveLock* toggles body-locked (walking: panel distance 0.85 m; $\pm 30^\circ$ H, $\pm 20^\circ$ V) and world-locked (stationary > 0.5 s, within 1 m radius) to balance accessibility and predictability. Shifts out of view during rapid head turns to avoid tunneling, reverting to body-lock beyond range [40].

AdaptiveFlow (Fig. 1d): *AdaptiveFlow* smoothly transitions between body-locked (slow motion: panel distance 0.85 m) and temporarily world-locked (fast walking/turns) via 2-second low-pass filtering ($\alpha \approx 0.6$ –0.8) to avoid snaps. Clears central FOV for stability [2], easing back fluidly.

While *AdaptiveLock* applies world-locking when stationary to prioritize spatial predictability, *AdaptiveFlow* uses brief world-locked breaks during continuous body-locked access to preserve temporal availability.

5.2 Implementation Platform

Our system treats anchoring as an automated attentional policy that responds to real-time motion cues rather than requiring manual user adjustment. All four techniques were implemented in Unity 2022.3 using MRTK 2.8 on the Microsoft HoloLens2 [29]. The HMD provides a 2048×1080 resolution per eye, a 60 Hz refresh rate, and a 52° diagonal field of view. To support gaze-contingent analysis, we utilized the device’s integrated 30 Hz eye-tracking sensors. The system calculates per-frame updates (60 Hz) for the UI panel’s position, orientation, and scale by processing 6-DOF head-tracking and body transform data, ensuring that adaptive transitions remain synchronized with the user’s physical movement. It logs eye-tracking, system events, and user interactions at 10 Hz.

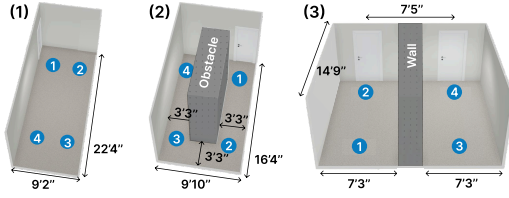
Specifically, the pipeline computes walking speed, head-turn velocity, and user–panel distance from the camera’s 3D trajectory and quaternion rotation. These kinematic variables serve as parameters for Unity solvers, which interpolate the transitions between anchoring states. While each technique employs a unique mapping between motion signals and anchoring behavior, they all use the same sensing and processing architecture. Full implementation details and source code are available in the [anonymized repository](#).

6 Study 1: INTERFACE PERFORMANCE UNDER BASIC MOBILITY DEMANDS

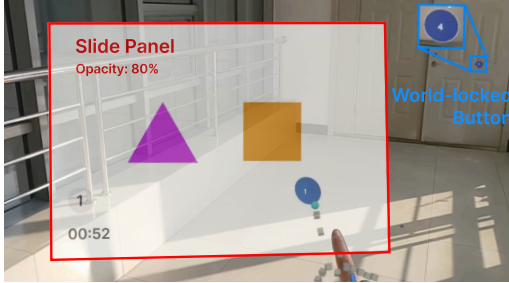
6.1 Motivation and Research Questions

Study 1 investigated interface performance during controlled mobility paired with basic information processing. We examined how *technique* influence individual performance in dual-task scenarios that demand spatial navigation alongside information retention—a typical setup where users balance environmental awareness and interface access. With simple waypoint navigation and moderate memory recall demands, we predicted that digital information accessibility would be the primary factor affecting memory performance. We also expected trade-offs: accessibility gains from adaptive techniques might introduce cognitive costs during transitions. The task design emphasized continuous movement between spatially distributed waypoints, creating relatively high mobility demands where users frequently needed to access digital information while walking between locations.

RQ1: *How do AR information anchoring techniques affect user performance and experience in multitasking with moderate cognitive and relatively high mobility demands?*



(a) Experimental environments: (1) *Open Space*, (2) *Narrow Corridor*, and (3) *Room-to-Room*, detailing dimensions and the four fixed spatial waypoints (1–4).



(b) A first-person view showing the execution of the two tasks, along with Slide panel. Figure 2: Study 1’s environmental layouts and experimental tasks.

6.2 Method

Participants. We recruited 12 volunteers (P1-P12; 10 male, 2 female; age 21–30, $M=24.3$, detailed in Sec. 3) to take part in the study.

Apparatus. As mentioned in Sec. 5.2, the AR application ran on HoloLens2, displaying a semi-transparent slide panel (Fig. 2b). Locomotion states segmented walking vs. stationary periods, yielding Area-of-Interest (AOI)-based gaze metrics (dwell time, fixation duration in Slide panel and Other AOIs) and attention switches. Post-experiment questionnaires used a desktop PC, keyboard, and mouse.

6.2.1 Environment Setup

The study occurred in a quiet lab with controlled indoor lighting. As shown in Fig. 2a, we tested three room configurations—*Open Space*, *Narrow Corridor*, and *Room-to-Room*—varying visibility, occlusion, and mobility to evaluate anchoring techniques (detailed below). Participants walked between four random app-defined waypoints in each, facing distinct panel visibility and locomotion constraints.

Open Space (Fig. 2a(1)): Low-complexity control with unobstructed sightlines and paths, minimizing navigation costs and maximizing attention for the cognitive task.

Narrow Corridor (Fig. 2a(2)): Central barrier forces users close to walls, requiring switches between body-centered and environment-centered views to keep the AR panel visible while staying aware of side boundaries.

Room-to-Room (Fig. 2a(3)): Induces re-engagement via doors and hallways, creating ‘event boundaries’ [5]. Tests anchoring robustness against doorway negotiation, stopping, and panel re-acquisition after locomotion interruptions.

6.2.2 Task Design

Participants completed two **concurrent** tasks to mimic navigating while processing information (e.g., collaborate on a shared artifact).

Mobility Task: Waypoint Click. In the mobility task, four fixed waypoints (labeled 1–4), as shown in Fig. 2a, appeared six times in random order at the bottom of a slide over each 5-minute condition (50 s per interval). Participants walked to the corresponding waypoint via any route and confirmed arrival by clicking a world-locked blue button there. This design created sustained mobility

demands as participants continuously moved between distributed locations throughout the session, similar to prior AR walking and waypoint-based navigation tasks [42].

Cognitive Processing Task: Slides Memory. To simulate remembering short-term history (e.g., recently presented task-relevant information during AR information processing), we developed a memory task similar to prior work [44, 16]. The cognitive task involved viewing a six-slide deck (50 s per slide) showing two items each (Fig. 2b)—numbers, English words, or simple shapes—in a fixed sequence. After each condition, participants took a 12-item immediate-recall test (4 items per category) to isolate how anchoring techniques affect information encoding during active movement.

6.2.3 Procedure

Participants first completed eye-tracking calibration in the HMD, followed by a 5–10 minute training session. They then completed 12 experimental conditions in a 4 (*Technique*) $\times$ 3 (*Environment*) design. We counterbalanced the four techniques using a Latin Square but presented the environments in a fixed sequence—*Open Space*, *Narrow Corridor*, and *Room-to-Room*—with escalating mobility demands. This approach prioritized evaluating anchoring techniques while assessing performance across spatial contexts.

We grouped the 12 conditions into three sets, one per environment, with each set containing four trials (one per technique). Each condition lasted about 5 minutes and involved concurrent dual tasks, followed immediately by a questionnaire on task recall and subjective experience.

Participants took 5-minute breaks between conditions and 30-minute breaks between sets (with recalibration if needed). Sessions lasted 3–4.5 hours per participant and ended with a brief 10-minute interview.

6.2.4 Measures

We characterized visual attention using gaze-based measures capturing. **Interface Attention Ratio** quantifies how strongly participants allocated visual attention to the Slides Panel. For each participant and condition, we summed valid dwell time on the Slides Panel AOI and divided it by total valid gaze time (i.e., $\frac{T_{\text{Slides}}}{T_{\text{total valid}}}$). Higher values indicate stronger sustained allocation to the AR content [14].

Immediate Recall indicates participants’ ability to encode and remember slide content while multitasking. It was operationalized as *Immediate Recall* accuracy on a 12-item quiz administered after each condition. Each correct response earned one point, yielding a total score from 0 to 12.

We collected post-condition ratings to capture perceived anchoring quality, workload, usability, and preference, as subjective complements to behavioral measures. We use the Raw NASA-TLX (RTLX [13]) as a measure of overall task demand. We also administered a custom 7-point Likert questionnaire (1 = strongly disagree, 7 = strongly agree) assessing *perceived* temporal availability (e.g., always availability, steadiness) (Details in Appendix Sec. B-Tab. 2) [11, 37, 9] of anchoring techniques. For analysis, individual item ratings within each construct were averaged to produce a composite score for each participant and condition. We also collected overall usability using the System Usability Scale (SUS [3]) and *Preference Ranking*.

6.3 Results

Analysis. We conducted two-way repeated-measures ANOVAs for *technique* $\times$ *environment* measures, except for SUS, which used a one-way repeated-measures ANOVA (*technique* only). Preference rankings were summarized descriptively. Normality was assessed using Shapiro–Wilk tests; sphericity was evaluated using Mauchly’s test; and Greenhouse–Geisser

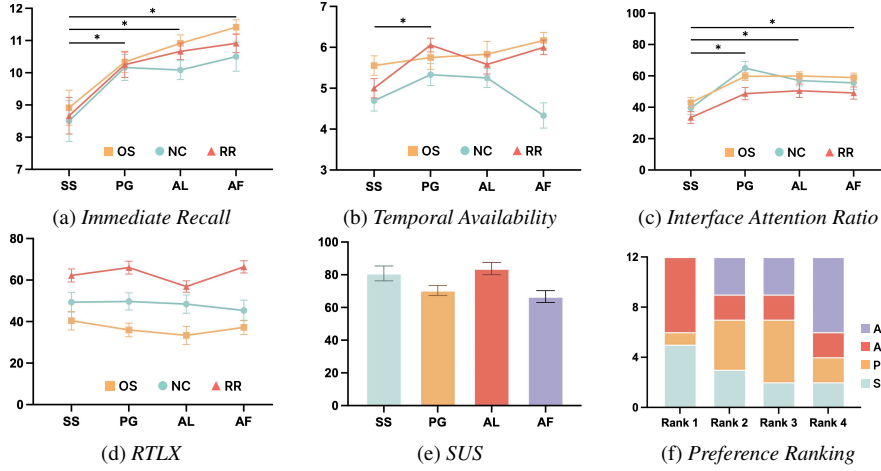


Figure 3: Study 1 results of *Technique* and *Environment*. * represents significant differences in *Technique* ($p < .05$). Error bars represent $\pm$ Standard Error of Mean (SEM). Here, *Technique*: SS = *ScaleStable*, PG = *PeriphGuide*, AL = *AdaptiveLock*, AF = *AdaptiveFlow*; *Environment*: NC = *Narrow Corridor*, OS = *Open Space*, RR = *Room-to-Room*. For details refer to Appendix Sec. C Tab. 3-Tab. 4.

corrections were applied when sphericity was violated. Post hoc pairwise comparisons used Bonferroni-corrected paired samples t -tests. Effect sizes are reported as partial eta-squared, interpreted following Cohen’s guidelines: small ($\eta_p^2 \approx .01$), medium ($\eta_p^2 \approx .06$), and large ($\eta_p^2 > .14$). Fig. 3 summarizes the results.

Immediate Recall (Fig. 3a). *ScaleStable* significantly impaired memory recall compared to all other techniques, independent of physical environment. We found a significant main effect of *technique* ($F_{3,33} = 17.79$, $p < .001$, $\eta_p^2 = .618$), with no significant *environment* or interaction effects (all $p > .05$). Post-hoc tests confirmed that *ScaleStable* (8.69) yielded significantly lower recall accuracy than *AdaptiveFlow* (10.94), *AdaptiveLock* (10.56), and *PeriphGuide* (10.25) (all $p < .030$). This pattern aligns with our theoretical expectation: the three alternative techniques support greater accessibility to digital information during movement, whereas *ScaleStable*’s world-locked positioning restricts interface access when users are mobile. This performance difference demonstrates that information accessibility directly shapes cognitive task performance during continuous mobility. Two complementary objective and subjective measures further validate this accessibility–performance relationship.

First, **Temporal Availability (Fig. 3b).** We identified significant main effects of *technique* ($F_{3,33} = 4.19$, $p = .013$, $\eta_p^2 = .276$) and *environment* ($F_{2,22} = 11.86$, $p < .001$, $\eta_p^2 = .519$), with no significant interaction ($p = .114$). Post-hoc tests showed that *ScaleStable* (5.08) was rated significantly lower in availability than *PeriphGuide* (5.71, $p = .034$). For *environment*, *Narrow Corridor* (4.9) was rated significantly lower in availability than both *Open Space* (6.8, $p = .003$) and *Room-to-Room* (5.6, $p = .010$).

Second, **Interface Attention Ratio (Fig. 3c).** The proportion of attention allocated to interfaces was independently influenced by *technique* ($F_{3,33} = 12.38$, $p < .001$, $\eta_p^2 = .530$) and *environment* ($F_{2,22} = 8.37$, $p = .002$, $\eta_p^2 = .432$), with no interaction ($p = .64$). Post-hoc tests revealed that a significantly smaller proportion of time was spent on interfaces in the *ScaleStable* condition (38.63%) than in all other conditions (*AdaptiveFlow*: 54.52%, *AdaptiveLock*: 55.85%, *PeriphGuide*: 57.79%; all $p < 0.008$). For *environment*, a similar pattern emerged: significantly less interface dwell time occurred in *Room-to-Room* (45.43%) than in *Narrow Corridor* (54.25%, $p = .004$) or *Open Space* (55.41%, $p = .021$).

Together, these results clarify why recall performance was poorest in the *ScaleStable* condition: its world-locked implementation results in lower perceived temporal availability of digital information, which in turn reduces visual attention and interface engage-

ment.

For **Workload (Fig. 3d)**, we found a significant main effect of *environment* ($F_{2,22} = 40.13$, $p < .001$, $\eta_p^2 = .785$), but no *technique* ($p = .255$) or interaction effects ($p = .771$). Post-hoc tests confirmed workload was significantly highest in *Room-to-Room* (62.88), intermediate in *Narrow Corridor* (48.22), and lowest in *Open Space* (36.74) (all pairwise $p \leq .030$). This suggests that spatial complexity, not interface design, drove cognitive demands.

However, a surprising **SUS (Fig. 3e)** pattern emerged that contradicted performance outcomes. *Technique* had a significant effect ($F_{3,33} = 3.58$, $p = .024$, $\eta_p^2 = .246$). Despite *ScaleStable*’s memory performance penalty, it ranked among users’ most preferred techniques: *AdaptiveLock* (SUS: 83.75, 50% first-place votes) and *ScaleStable* (SUS: 80.83, 42% first-place votes) topped usability ratings, capturing 92% of first-place preference votes. Users favored *AdaptiveLock*’s consistent access or *ScaleStable*’s predictable spatial reference, despite accessibility costs. *AdaptiveFlow* (66.67) and *PeriphGuide* (70.42), though more accessible, felt suboptimal due to ongoing field-of-view interference.

Qualitative feedback revealed why users preferred *technique* that avoided visual interference. Participants, like P3, described *PeriphGuide* as “always in the corner of my eye, distracting from navigation” and *AdaptiveFlow* as “constantly moving around, hard to ignore even when I didn’t need it.” by P8. In contrast, they appreciated *ScaleStable*’s approach “When I needed it, I knew exactly where to find it. When I didn’t, it wasn’t bothering me (P11).” *AdaptiveLock* struck an optimal balance: “Best of both worlds—available when I stopped, out of the way when I moved (P7).”

6.4 Study 1 Discussion

Study 1 demonstrated that during mobility with moderate cognitive demands, digital information accessibility directly enhances memory performance. However, users showed a strong preference against persistent field-of-view interference, even when it improved objective performance outcomes.

The results reveal two design strategies that emerged across conditions. *AdaptiveLock* achieved a strong balance by adapting interface placement to user state: body-locked when stationary to support information accessibility, and world-locked during movement to reduce interference. This design eliminated the disruptive 2-second transition present in *AdaptiveFlow*, yielding both the best objective performance and strongest user preference. As P10 described, it felt “like an extension of my body while I was moving, but [a] real object once I stopped.”

Notably, *ScaleStable* maintained high user preference despite substantial accessibility limitations and significantly lower memory performance. This dissociation suggests users strongly prioritize minimizing visual interference during movement, even when doing so reduces information access and impairs cognitive task performance.

Users valued *ScaleStable*'s stable spatial reference and complete lack of visual distraction, which supported a strong sense of control over their visual attention. Yet this preference came with a clear measurable cost: *ScaleStable* produced the lowest *Immediate Recall* (8.69/12) and *Interface Attention Ratio* (38.63%).

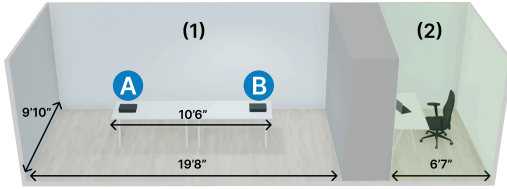
This dissociation between user preference and objective performance raises an intriguing question: How might these trade-offs shift under different task constraints? If environmental complexity changes or mobility demands vary, would the optimal balance between accessibility and interference shift toward different design solutions? *Study 2* investigates how these performance and preference patterns change under altered task conditions.

7 Study 2: INTERFACE PERFORMANCE UNDER COLLABORATIVE COGNITIVE LOAD

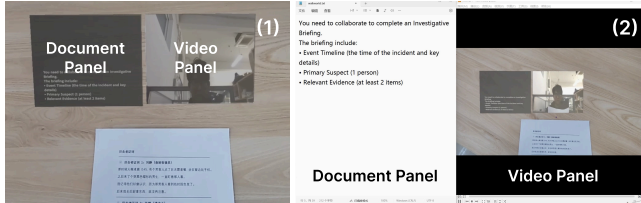
7.1 Motivation and Research Questions

Building on *Study 1*, *Study 2* tested *technique* under heightened environmental processing demands and collaborative cognitive load. This sensemaking task required constant attention shifts between detailed physical evidence inspection, digital information synthesis, and partner communication. Importantly, this study also reduced the proportional demand for mobility: while AR *wearers* moved between investigation stations, they spent significant time in stationary discussion and analysis with their remote partners. With these shifted demands, that complex environmental processing, collaborative reasoning, and reduced mobility emphasis, we investigated how the accessibility-stability trade-off would manifest.

RQ2: How does the accessibility-stability trade-off play out when AR interfaces support complex environmental processing and collaborative cognition?



(a) Two-room setup with a soundproof wall. (1) *wearer's* room. (2) *partner's* room.



(b) Dyad views. (1) *Wearer's* HMD view with document panel and partner video. (2) *Partner's* desktop view with shared document and streamed first-person AR view.

Figure 4: Study 2 physical setup and digital views.

7.2 Method

7.2.1 Participants

Twelve pairs ($N = 24$; 4 mixed-gender, 6 male, and 2 female dyads; details in Sec. 3) participated in the study. Each pair consisted of one *wearer* (Fig. 4a (1)), who navigated the physical environment while viewing the shared interface through an HMD (W1–W12; age 19–25, $M = 21.58$), and one *partner* (Fig. 4a(2)), who joined remotely from a fixed location using a desktop (P1–P12; age 20–27,

$M = 23.0$). All participants reported at least five years of experience with conventional online meetings (e.g., video conferencing). All *wearers* had no prior experience with AR meeting applications.

7.2.2 Apparatus

This study employed an asymmetric AR–desktop setup (Fig. 4) to support synchronous dyadic collaboration, building upon established paradigms for mixed-reality cross-view teamwork [32, 23, 7, 22].

The AR hardware and system parameters were identical to *Study 1* (Sec. 5.2), utilizing a HoloLens 2 with the same interface layout, anchoring techniques, and motion thresholds. As shown in Fig. 4b(1), the *wearer's* interface comprised a live video window of the *partner* alongside a shared document panel, which synchronized with the *partner's* real-time documents via Windows Internet Information Services (IIS) over WiFi. Both panels were governed by the assigned anchoring condition. To maintain experimental control, the document panel maintained a fixed size and prohibited manual scrolling or repositioning. For analysis, gaze data were mapped to three predefined AOIs: the Document Panel, the Video Panel, and the Environment/Other (Fig. 4b(1)).

The *partner* used a laptop (Honor X16, Intel Core i7-12700H, 16GB RAM) to monitor the *wearer's* first-person perspective via a Real-Time Streaming Protocol (RTSP) feed at 720p 30fps (≈ 100 ms latency) using VLC media player.

7.2.3 Task Design

To simulate real-world collaboration, we designed a role-playing task inspired by collaborative sense-making methodologies [19]. Dyads participated in a murder-mystery investigation requiring continuous information exchange and physical movement. Specifically, the *wearer* and *partner* pooled asymmetric evidence to identify a single correct suspect.

As illustrated in Fig. 4a, the *wearer* navigated between two document stations (A and B) positioned 10'6" ft apart at opposite ends of a central table. Station A contained witness statements, whereas Station B held physical crime scene evidence. To maintain spatial predictability, the *wearer* returned all materials to their original stations after inspection. Concurrently, the remote *partner* worked at the desktop station, reviewing suspect profiles and case backgrounds.

This asymmetric distribution of information [15] forced the *wearer* to continuously switch attention among the physical environment, the shared AR interface, and the remote *partner*. Together, the dyad consolidated their findings into a shared *Reasoning Brief*, which required: (1) an event timeline, (2) a primary suspect, (3) two supporting pieces of evidence, and (4) a final conclusion. To prevent learning effects across conditions, we generated five structurally identical investigation scripts using ChatGPT [31], which were subsequently manually edited to ensure uniform length, complexity, and difficulty.

7.3 Procedure

Participants were briefed on the study objectives and randomly assigned to either the *wearer* or *partner* role. The *wearer* received 5–10 minutes of task training and practiced interacting with physical materials at the document stations, followed by HMD fitting and eye-tracking calibration. Simultaneously, the *partner* received separate training on utilizing the shared document interface for real-time note-taking.

Employing a within-subjects design, each dyad completed four untimed investigation sessions—one for each anchoring technique—counterbalanced using a Latin Square. Each session progressed as follows: (1) the *partner* read the case background aloud; (2) both participants initiated the investigation with unrestricted verbal communication; (3) the *wearer* navigated between

Stations A and B to inspect physical evidence, while the *partner* reviewed digital suspect materials and populated the shared Reasoning Brief; and (4) the session concluded once both participants agreed the Brief was complete. Immediately after each session, participants completed subjective experience questionnaires and took a compulsory 15-minute break.

After all four sessions, the dyad was interviewed separately in a 10-minute semi-structured interview to explore how the different anchoring techniques affected their collaborative process. The entire experiment lasted approximately 3-3.5 hours per dyad. An experimenter monitored the system from an adjacent control room, intervening exclusively for technical issues.

7.3.1 Measures

Similar to *Study 1* (Sec. 6.2.4), we measured *Interface Attention Ratio*, which quantifies the proportion of visual attention allocated to the meeting interface, consisting of the Video panel and the Document panel (i.e., $\frac{T_{\text{Video}} + T_{\text{Document}}}{T_{\text{total valid}}}$). We also collected measures from *wearer* including *RTLX*, and perceived *Spatial Predictability* (e.g., predictability/direction/position/relocatability of panels; 1 = strongly disagree, 7 = strongly agree; Details in Appendix Sec. B-Tab. 2 [11, 37, 9]). For *RTLX*, we also collected *partner's* responses and averaged them per dyad to obtain a single score.

We measured collaboration quality from two complementary perspectives: subjective evaluation and objective performance. Subjective collaboration quality was captured by *Perceived Collaboration*, computed by averaging both team members' independent ratings (1 = strongly disagree, 7 = strongly agree) to the post-session statement: "Our team produced a high-quality reasoning brief with well-supported conclusions," yielding one score per dyad per condition. Objective collaboration performance was measured using task completion time and murder mystery accuracy, based on the correctness of the team's final conclusions, with correct conclusions scored as 1 and incorrect conclusions scored as 0.

7.4 Results

We followed the same analysis procedure (Sec. 6.3) as in *Study 1*, except that we used one-way repeated-measures ANOVAs.

Fig. 5 summarizes the results. All dyads successfully completed the task across all 48 sessions.

Task performance showed no significant differences among the four techniques in completion time or murder mystery accuracy (average time: 22.96 minutes, SD = 7.24; average accuracy: 0.60, SD = 0.49). However, collaboration experience varied significantly. For *Perceived Collaboration*, *technique* had a significant large effect, $F(3, 33) = 3.84$, $p = .018$, $\eta_p^2 = .259$. *ScaleStable* (6.52) received significantly higher ratings than *AdaptiveFlow* (6.30, $p = .044$) and *PeriphGuide* (6.32, $p = .004$), while *AdaptiveLock* (6.40) did not differ significantly from the others.

Post-study rankings mirrored these findings: *ScaleStable* (58% (7/12)) and *AdaptiveLock* (42% (5/12)) captured all first-place votes, with no last-place votes for *ScaleStable*. *PeriphGuide* ranked intermediately, while *AdaptiveFlow* was consistently least favored (67% (8/12) last-place votes).

Several metrics explain *ScaleStable's* superior perceived collaboration quality and high preference. First, *Spatial Predictability* showed a significant large effect, with *ScaleStable* rated significantly higher than *AdaptiveFlow*; other pairwise comparisons were nonsignificant. Second, *Interface Attention Ratio* also revealed a significant large effect. Wearers spent significantly less time on the interface with *ScaleStable* than with *AdaptiveFlow* or *PeriphGuide* (*AdaptiveLock* differed only marginally from *AdaptiveFlow*). This reduced interface attention did not impair task performance or collaboration quality. Instead, under high environmental processing demands, it appears preferable to minimize interface attention while

preserving access—freeing users for focused thinking and evidence analysis rather than chasing a moving panel.

RTLX workload explains *AdaptiveLock's* second-place preference: technique had a significant large effect, with *AdaptiveLock* significantly less demanding than *AdaptiveFlow*; other comparisons were nonsignificant. *ScaleStable* and *AdaptiveLock* were the two lowest-workload techniques overall, indicating that continuous adaptation disrupted users more than discrete mode switching under high cognitive load.

Interviews reinforced these preferences. Participants praised *ScaleStable's* predictability for collaboration: "The fixed location felt stable, and I always knew where it was (W11)"; "Easier to discuss, stayed aligned on the document (W6)". *AdaptiveFlow* frustrated under high load: "Panel always on the move—chasing it with my eyes when I needed to focus on evidence (W5)". *PeriphGuide* interfered persistently: "Always in my way when trying to examine evidence carefully—couldn't focus properly (W8)". *AdaptiveLock* struck a balance: "Less tiring than the moving one, easier to manage when I needed to focus on evidence (W10)".

7.5 Study 2 Discussion

Study 2 demonstrated that when task complexity increased and mobility demands decreased, the optimal anchoring strategy shifted away from prioritizing accessibility and toward minimizing field-of-view interference. Unlike *Study 1*, where *AdaptiveLock* benefited mobile multitasking, *ScaleStable* emerged as the technique with both the highest perceived collaboration quality and the highest user preference during collaborative sensemaking.

The results suggest several possible factors for this shift. First, the complex environmental inspection task likely made field-of-view interference more costly. Participants needed more visual attention to examine physical evidence and interpret spatial relationships. Persistent overlays (*PeriphGuide*) and moving panels (*AdaptiveFlow*) therefore directly competed with this task-relevant perception. In this context, *ScaleStable* supported performance by remaining spatially predictable and visually unobtrusive. As W6 noted, "It was easier to discuss, and we stayed aligned on the document." This indicates that stable world-locked placement reduced interface management demands and better supported controlled attention.

In addition, the nature of collaborative reasoning altered the accessibility-interference trade-off in two ways. Compared with *Study 1*, participants in *Study 2* spent more time stationary while discussing and coordinating. This reduced mobility meant there was less time in which the world-locked interface would be unavailable, diminishing the primary advantage of *AdaptiveLock*. Furthermore, the task required more focused thinking and analysis. The mode-switching inherent in *AdaptiveLock*—between a head-locked state for immediate access and a world-locked state for stability—became more cognitively distracting than maintaining a single, predictable world-locked mode as in *ScaleStable*. This helps explaining why *AdaptiveLock*, which performed best in *Study 1*, no longer provided the same advantage, while *ScaleStable* became more preferred. Conversely, *AdaptiveFlow* again produced high gaze engagement but poor performance outcomes, suggesting that increased panel tracking reflected distracting interface-management effort rather than productive coordination [28].

Together, these findings demonstrate that anchoring strategies optimized for one type of task may not transfer directly to another. When environmental engagement, mobility demands, and task complexity change, the relative weights of field-of-view interference and information accessibility shift, resulting in different techniques rising to the forefront.

Notably, we observed that similar to *Study 1*, users disliked the constant field-of-view interference of *PeriphGuide*, and the interference caused by the two-second transition in *AdaptiveFlow* in both

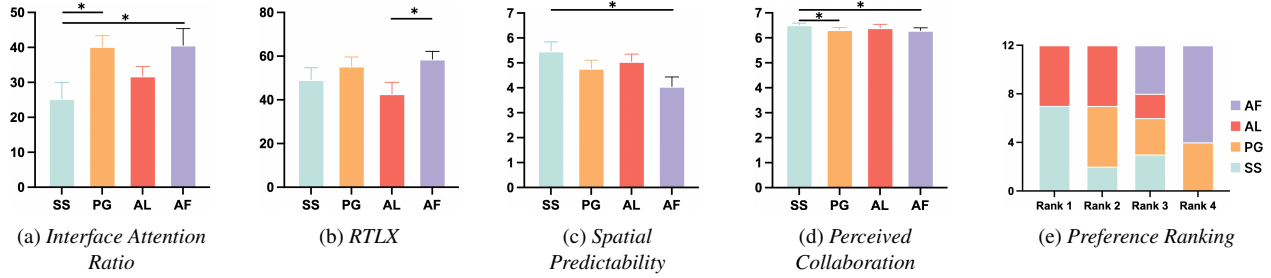


Figure 5: Study 2 results of *Technique*. * represents significant ($p < .05$). Error bars represent $\pm$ SEM. Here, *Technique*: SS = *ScaleStable*, PG = *PeriphGuide*, AL = *AdaptiveLock*, AF = *AdaptiveFlow*. For details refer to Appendix Sec. C-Tab. 5.

tasks. This suggests that when a significant amount of physical environment viewing is necessary, techniques that persistently or repeatedly intrude upon the user’s view may be less desirable.

8 GENERAL DISCUSSION

8.1 The Dual-Attention Challenge in Mobile AR

Our findings reveal an important tension in mobile AR interface design: users often need to simultaneously manage environmental processing (spatial navigation, checking physical entities/obstacles, real-world task execution) and virtual content access (information retrieval, interface interaction, digital task completion) while navigating dynamic visual fields. This dual-attention demand becomes particularly complex during mobility, as the relative positioning of both environmental and virtual elements continuously changes, requiring frequent visual-spatial updating and attention reallocation between competing information sources.

8.2 The Interference-Accessibility Trade-off

Field-of-View Interference vs. Information Access In mobile AR usage scenarios, digital interfaces inherently create environmental viewing interference. Body-locked and head-locked interfaces provide immediate virtual access but impose persistent visual occlusion that becomes costly when environmental processing demands are frequent and detailed. World-locked positioning with dedicated spatial placement addresses this visual competition by allowing users to consciously control when to shift visual focus based on task priorities.

However, world-locked approaches face accessibility challenges during mobility. Content clarity degrades with viewing distance and angle, and interfaces may become completely invisible when users move beyond optimal viewing ranges. This creates a design tension between minimizing interference and maintaining access.

Adaptive Solutions and Their Hidden Costs Combining world-locked and body-locked approaches appears to resolve this trade-off through context-sensitive positioning: world-locked during stationary periods (reducing interference while maintaining accessibility) and body-locked during movement (solving mobility access problems). However, our findings reveal that such adaptive approaches can potentially introduce cognitive disruption costs. Each spatial transition requires users to reestablish visual strategies while maintaining complex task states, imposing cognitive overhead that can outweigh accessibility benefits. Single-mode techniques (e.g., *ScaleStable*) can better support sustained cognitive performance by maintaining consistent spatial-visual expectations, allowing users to develop automatic interface access patterns without disrupting higher-level reasoning processes.

8.3 Context-Dependent Optimization

Our studies demonstrate that optimal AR anchoring design requires task-specific visual-cognitive balance rather than universal accessibility solutions. The relative importance of interference minimization versus information access shifts based on contextual demands:

High Environmental Processing Demands: When tasks require detailed visual examination or complex spatial reasoning, minimizing field-of-view interference becomes more important. Field-of-view interference can impair environmental processing quality, making world-locked approaches with minimal interference more desirable despite some accessibility limitations.

High Mobility Demands: When continuous movement dominates task requirements, accessibility problems become more significant, creating greater value for techniques that maintain information access during motion. However, any accessibility solution must carefully balance benefits against the costs of persistent field-of-view interference with navigation and environmental awareness.

High Cognitive Load Scenarios: During complex reasoning, collaboration, or sensemaking tasks, cognitive disruption costs become more expensive. Users benefit from predictable, stable interface positioning that supports sustained focus rather than adaptive techniques that require ongoing attention management, even when such adaptation theoretically optimizes an individual factor.

8.4 Design Implications: Context-Aware Interface Strategies

These findings suggest that future AR systems may need to consider dynamic anchoring strategies that optimize the interference-accessibility-disruption balance based on real-time task assessment rather than motion-based heuristics alone. Effective context-aware systems can consider:

Environmental Complexity Assessment: Monitor visual clutter, spatial navigation demands, and environmental processing requirements to determine interference tolerance levels. Higher environmental complexity increases the cost of visual occlusion, favoring world-locked positioning.

Cognitive Load Detection: Assess reasoning depth, multitasking demands, and collaboration requirements to evaluate disruption costs. Higher cognitive loads favor stable, single-mode techniques over adaptive approaches that impose transition overhead.

Mobility Pattern Analysis: Track movement frequency, duration, and spatial patterns to determine accessibility needs. Sustained mobility periods increase the value of body-locked access, while mixed mobility-stationary patterns may benefit from carefully implemented adaptive approaches.

Task Phase Recognition: Identify when users are in focused processing modes (detailed examination, complex reasoning, collaborative discussion) versus active transition modes (navigation, information gathering). Stable positioning during focused phases with selective adaptation during transitions can minimize disruption while maintaining necessary access.

User Attention State Monitoring: Detect when users are engaged in sustained environmental processing, virtual content access, or rapid attention switching to optimize interface positioning for current attention allocation patterns.

This context-aware approach suggests that optimal AR interface design lies not in perfecting individual anchoring techniques, but in developing intelligent systems that can dynamically select and

configure anchoring strategies based on holistic assessment of user needs, environmental demands, and cognitive state.

9 LIMITATIONS AND FUTURE WORK

Our findings are limited by a tech-savvy participant pool, reliance on immediate recall in *Study 1*, and the brief (30-minute) dyadic sessions in *Study 2*. Future research should evaluate these techniques with diverse populations, larger groups, and extended durations across varied AR domains. Exploring multimodal approaches (e.g., spatial audio, haptics) for spatial awareness could further reduce visual attentional costs. Finally, future work must distinguish productive gaze coordination from interface struggle and explore dynamically adapting anchoring based on context and task demands.

10 CONCLUSION

Our research demonstrates that effective mobile AR anchor design requires contextual optimization of the visual-cognitive balance, rather than one-size-fits-all accessibility solutions. In two studies varying task demands and mobility patterns, we show how optimal anchoring strategies shift with environmental processing complexity, cognitive load, and movement needs.

Our key contribution pinpoints when and why design priorities should shift: field-of-view interference costs can outweigh accessibility benefits during tasks needing sustained environmental processing, while cognitive disruptions from adaptive strategies can exceed their value amid complex reasoning and collaboration. These findings provide empirical support for expanding AR anchor design considerations.

Designers should optimize the full visual-cognitive system holistically to handle dual-domain information processing—moving beyond isolated factors like accessibility, field-of-view interference, or simple motion heuristics toward intelligent systems attuned to dynamic environments, cognitive demands, and user context. Future AR interfaces should reframe the question: not “How can we make digital content more accessible?” but “How can we best support users’ complete visual-cognitive workflow?”

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